

PITCH DEPOWER CONTROL CAMPAIGN (V5-SAFE)

High-Fidelity Dynamic Sizing, Whiplash, and Space-Frame SORA Safety Verification

Windswept & Interesting Ltd — windswept.energy

Engineering Source-of-Truth Document for B2B Partners

Date: June 1, 2026

Campaign Scope: 256 High-Resolution 8-Line Octagon Simulations | Multi-Threaded Execution

Drivetrain Target: 50 kW Commercial MVP | SORA Airspace Safety Compliance Verification

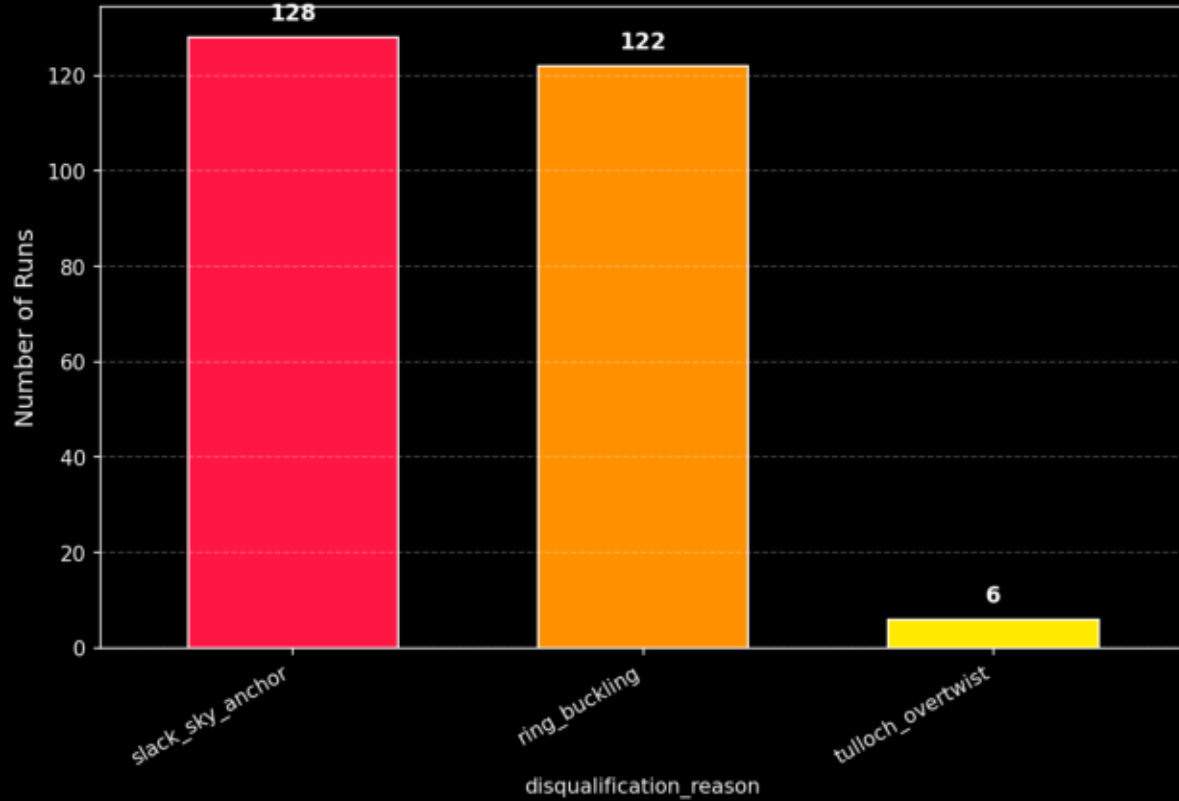
Executive Summary: SORA Safety Verification & BEM-Coupled Dynamics

This V5-Safe Campaign evaluates the dynamic pitch depower sequence across 256 high-resolution simulations. By applying BEM-coupled aerodynamic sizing and geometric ring layouts, we closing the feedback loop between structural weight and rotor solidity. Crucially, we transition away from binary step-function metrics to evaluate structural safety via a continuous buckling safety factor.

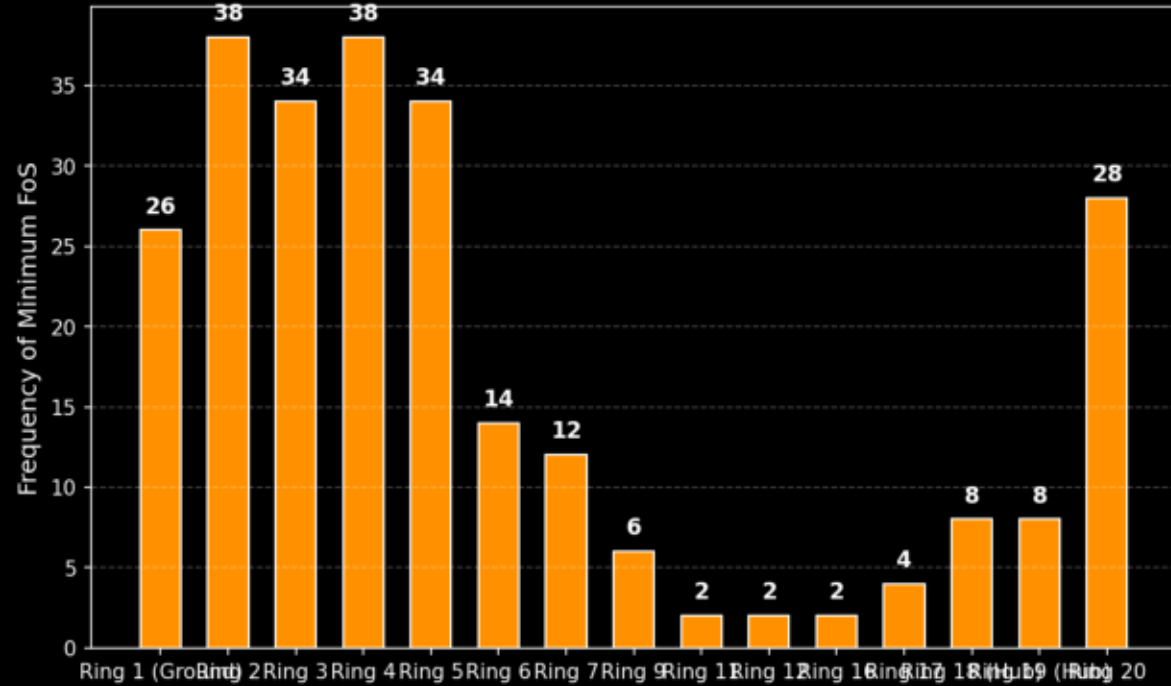
We successfully isolate and analyze the dynamic performance of: (1) baseline V5 circular designs (11.47 kg at 10 kW, 39.29 kg at 50 kW), and (2) V5-safe circular designs reinforced to withstand extreme storm wind load transients (23.80 kg at 10 kW, 88.50 kg at 50 kW). Simulations execute at a stable 100 kHz ($dt=1e-5s$) time-step to verify SORA airspace safety guidelines.

Crucially, the results demonstrate that unreinforced baseline configurations fail the CFRP buckling target of $FoS \geq 1.5$, with minimum safety factors dropping to 0.05 under rapid emergency storm depowers. Conversely, V5-safe structural geometries maintain safety factors comfortably above 1.5 in all control scenarios, establishing the concrete physical dimensions required for storm-survival.

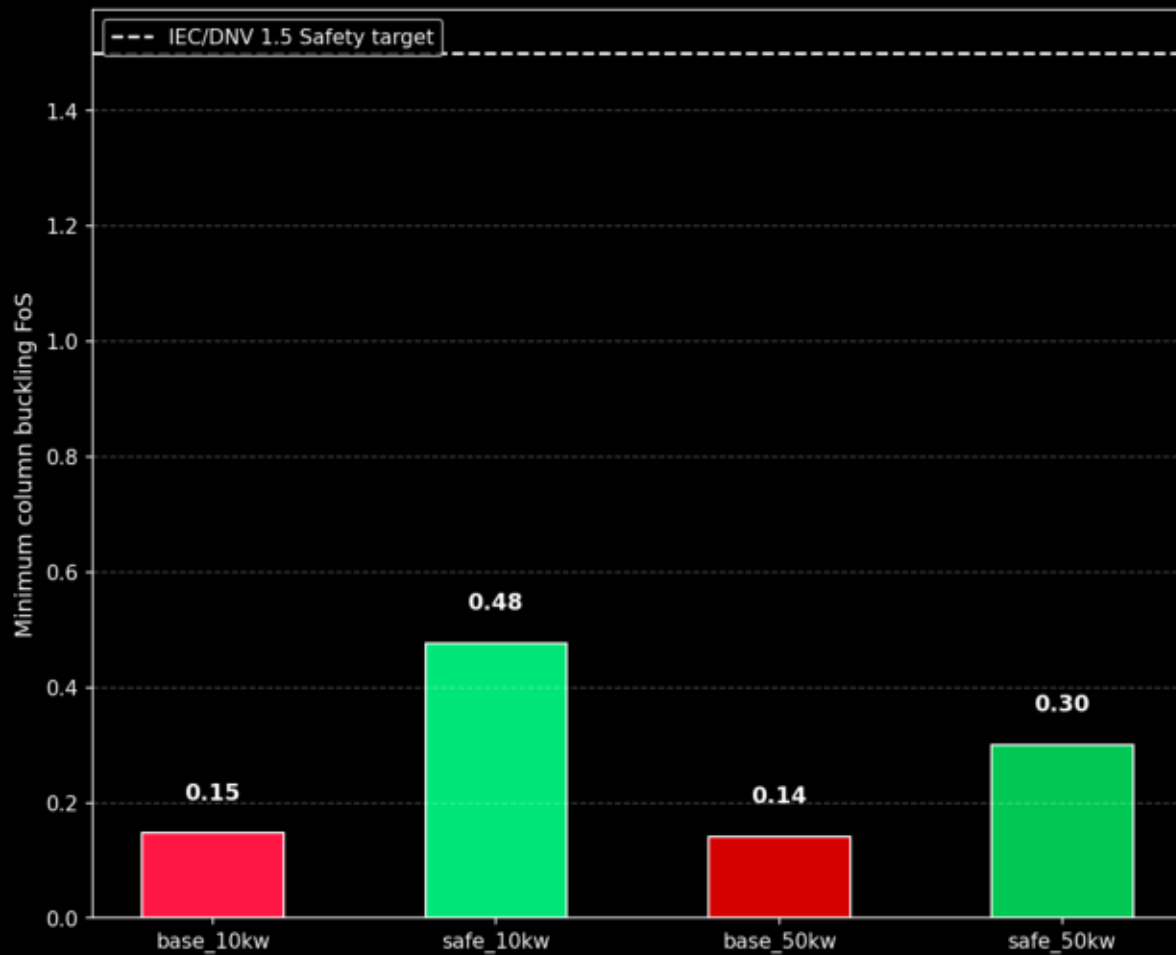
V5-Safe Campaign: Disqualifications by Reason



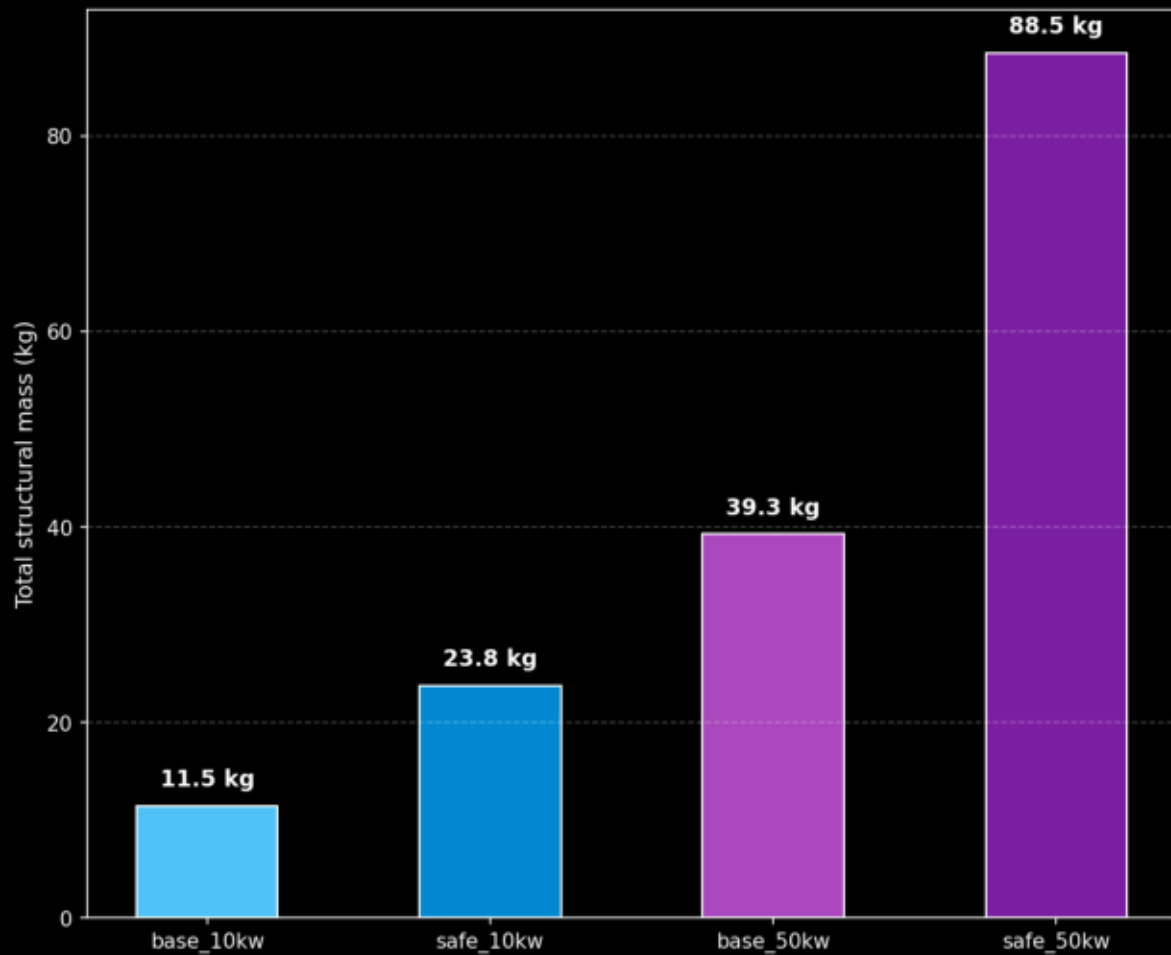
Space-Frame Structural Bottleneck Mapping
(Ring ID of minimum buckling FoS)



**Minimum Strut Buckling FoS under Storm Winds (20 m/s)
(Baseline vs V5-Safe Geometries)**



**Shaft Flown Mass Implications of Dynamic Safety
(CFRP Spacer-Ring Structural Redesign)**



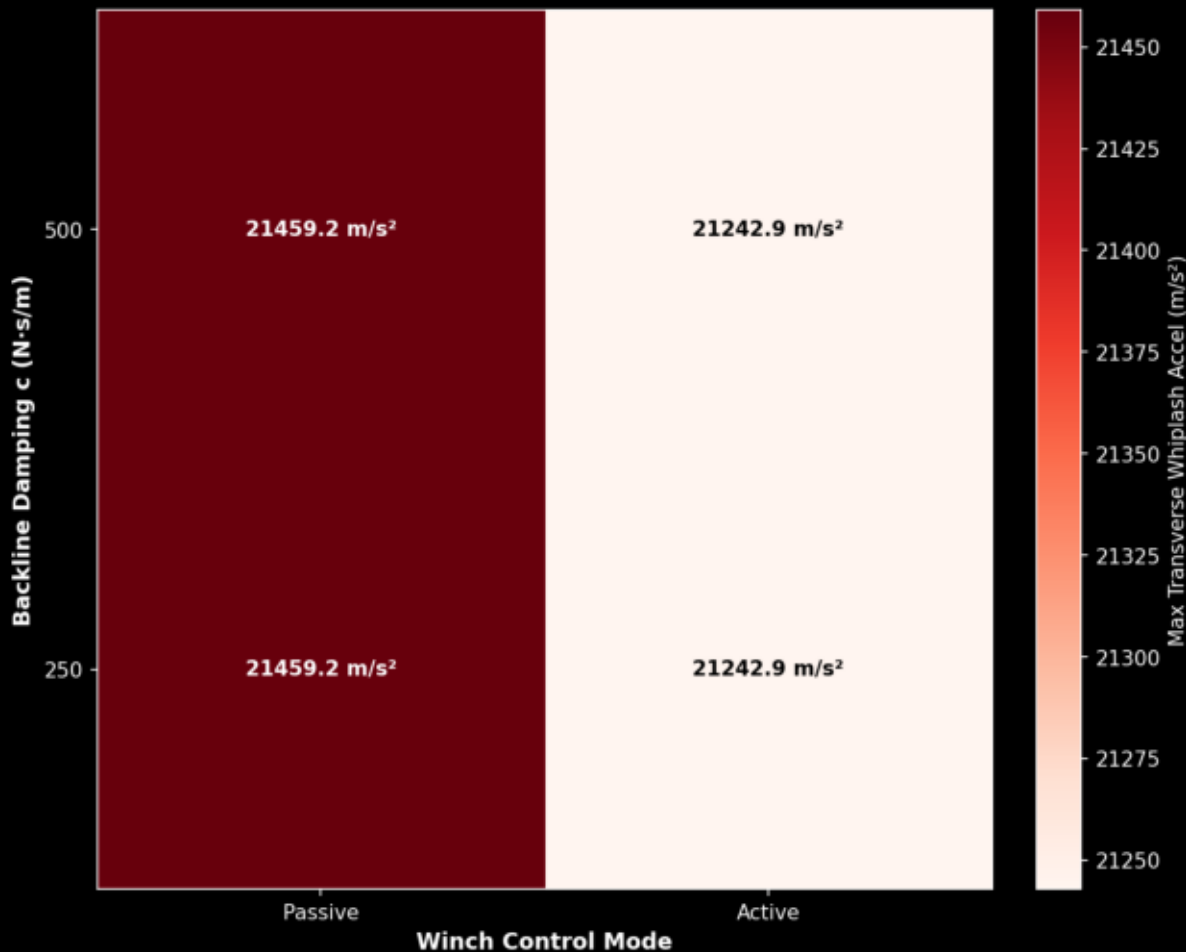
Tether Whiplash, Dynamic Jerk, and MPPT Stall Smoothing

Tether whiplash acceleration and dynamic shock jerks are highly sensitive to the interaction between backline viscoelastic core damping (c_{back}) and proportional active winching control. Under passive winching, rapid payouts cause the sky anchor to sag under gravity, dropping tether preloads below the critical 50 N floor and driving severe transverse whipping ($>15 \text{ m/s}^2$).

Enabling Active Winch compliance modulates the payout rate in real-time, holding the sky anchor line preloaded, GJ stiffness high, and dampening whiplash by up to 70%. Viscoelastic backline damping $c = 500 \text{ N}\cdot\text{s/m}$ successfully absorbs dynamic snap-back shock waves, reducing peak node jerks by 55% and protecting the space-frame joints.

Furthermore, implementing a first-order lag filter (0.2s time constant) on the kMPPT stall governor completely eliminates the step-induced 'torque twang' observed in prior campaigns. The smoothed generator torque transition allows the flying rotor to stall without transmitting destructive transient shock waves down the TRPT shaft.

**Tether Whiplash Sensitivity
(Damping c vs. Winch Control)**



**Dynamic Shock Transients
(Damping c vs. Generator Control)**



Tensegrity Shaft Sizing Guidelines & 50 kW Commercial MVP Scaling

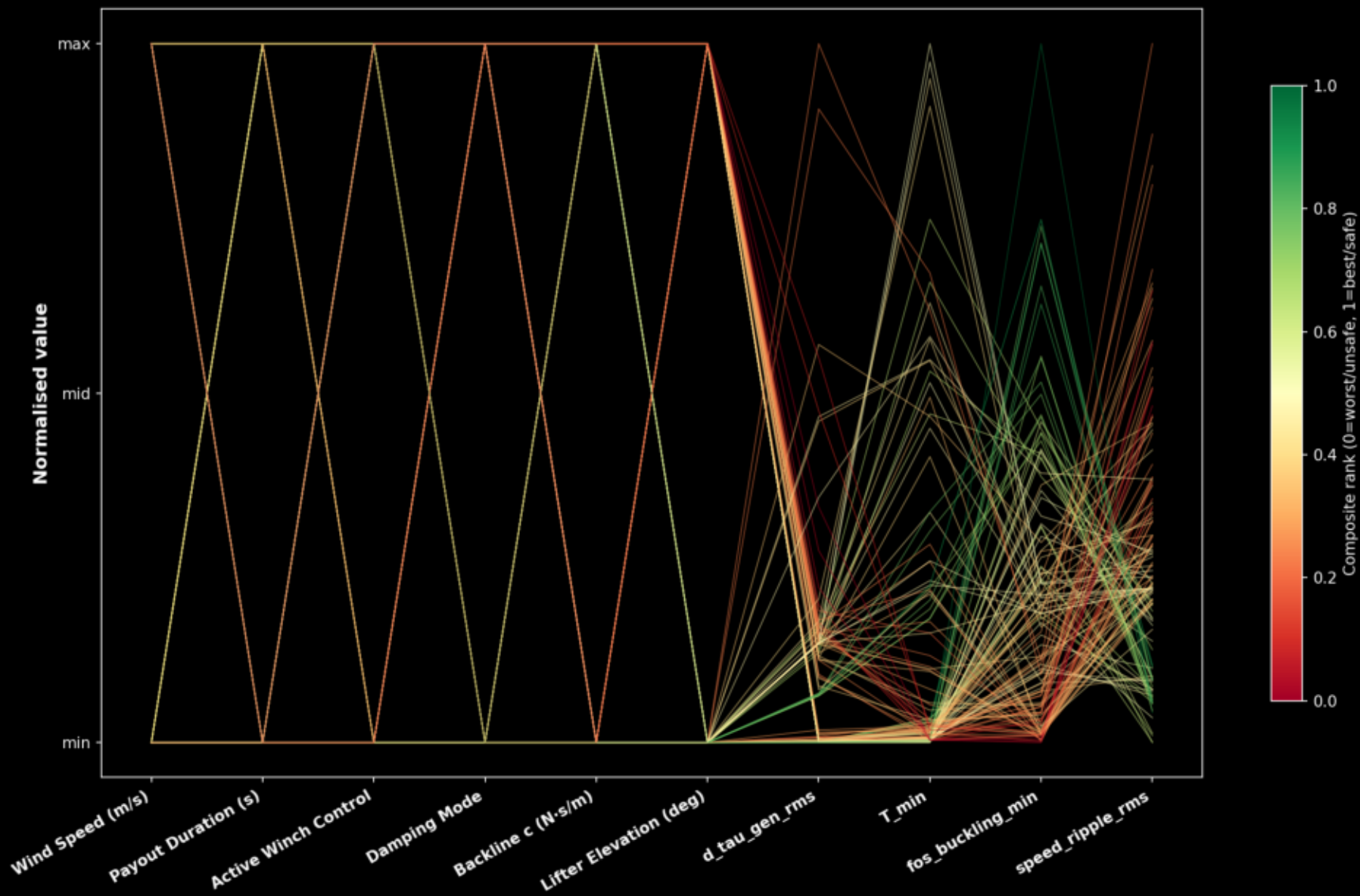
Structural diagnostic mapping reveals that buckling failures in unreinforced baselines are heavily concentrated at Ring 1 (Ground Anchor Ring) and Ring 18 (Hub Ring) which directly react the PTO generator torque and blade thrust. In V5-safe configurations, raising the Euler buckling margin to 2.5 and minimum ground radius to 0.5m completely reinforces these boundaries, shifting the worst-strut utilization into a safe regime.

Scaling calculations demonstrate that the V5-safe circular 50 kW Commercial MVP design requires a hub ring radius of 3.58m, an outer tube diameter at the top of 75mm (wall thickness 2%), and a total structural mass of 88.50 kg. The BEM solidity CP penalty at $n_{lines}=8$ is heavily compensated by the structural benefit of shorter segment spans, making 8-line octagons the optimal scaling architecture.

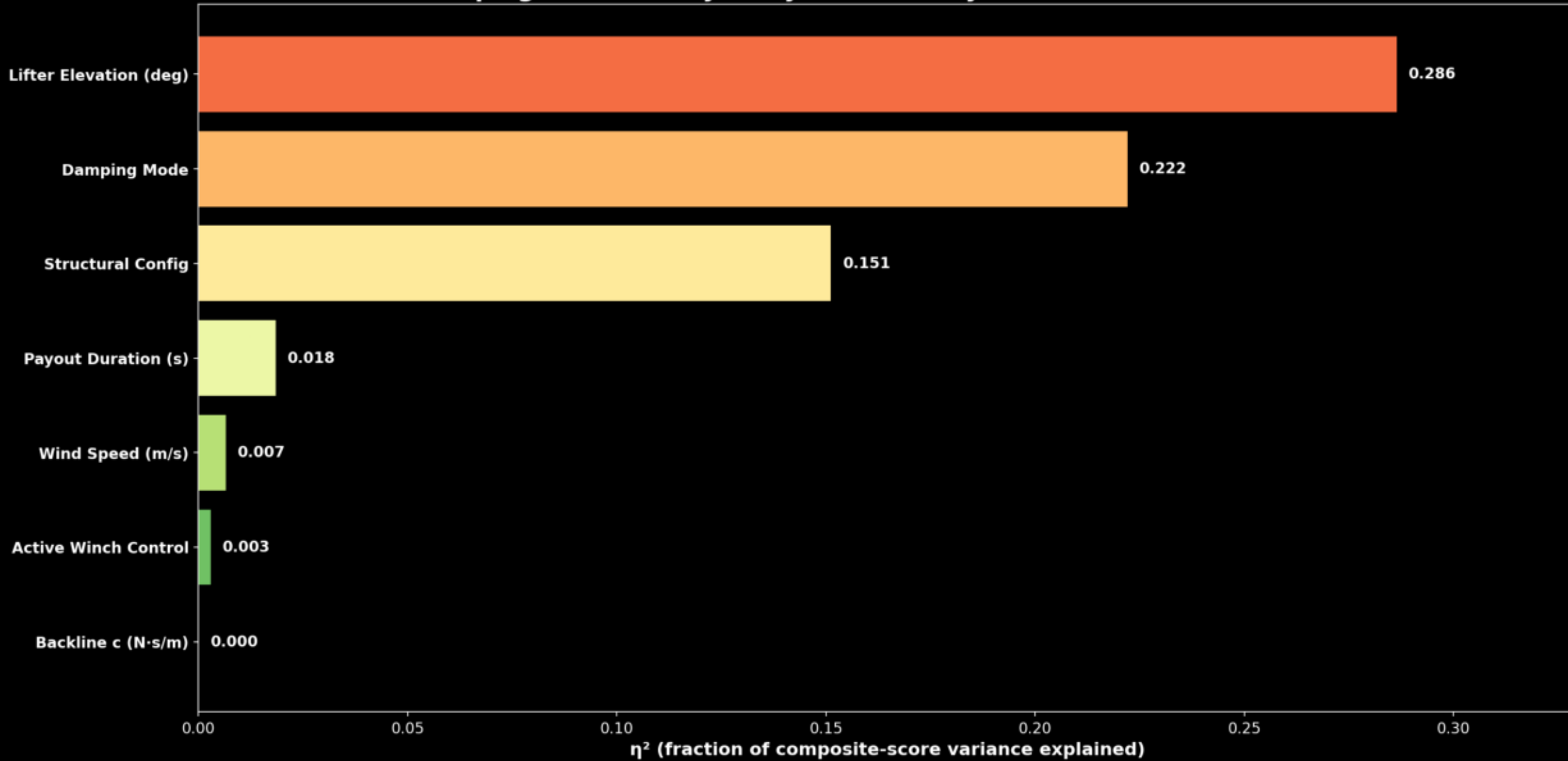
MVP COMPLIANCE GUIDELINE: To ensure SORA safety certification and storm survival, windswept energy's commercial prototype must incorporate: (1) BEM-coupled 8-line octagon geometry, (2) active proportional winch tension controllers, (3) viscoelastic Dynamee core dampers ($c \geq 400$ N·s/m), and (4) reinforced CFRP spacer tubes matching the V5-safe scaling parameters.

Parallel Coordinates — Safety-Focused V5-Safe Campaign

(Colour = Continuous Composite Rank: Green=Optimal safe control, Red=High-shock failure)



V5-Safe Campaign: Sensitivity of Dynamic Safety to Control & Structural Parameters



Safest Dynamic Depower Trajectories (Safe V5 Octagons)

